

Too few, too many, or just right? Optimizing sample sizes for population-level inferences in animal tracking projects

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S1. Case study

This document describes a case study using the workflow presented in the main manuscript. Our goal was to obtain a reliable estimate (within a 5% error threshold) of both **mean home range area** and **mean movement speed** of a tagged population of African buffalos (*Syncerus caffer*) in a protected area. We wished to limit expenses by acquiring the minimum number of tags needed to obtain reliable mean estimates for the tagged population, while accounting for individual variation.

To inform our analyses, we used a GPS movement dataset of African buffalos tracked between 2005 and 2006 in Kruger National Park, South Africa (Cross et al., 2016), available on Movebank and with the ‘ctmm’ R package. We assessed range residency by examining the variograms of all six

individuals within this dataset. Variograms allow for a cursory confirmation of range residency by checking if the semivariance function reaches an asymptote (Calabrese et al., 2016; Fleming, Calabrese, et al., 2014; Silva et al., 2022). All but one (named "Gabs") fit the range residency assumption, so we extracted population-level species parameters using the remaining five individuals.

When choosing a suitable tag for our focal species, we adhered to the standard guideline that units should not exceed 5% of an animal's body mass. With an average African Buffalo weighing 567.5 kg, weight constraints are negligible as most commercially available GPS collars are well below the 5% body mass threshold. Given a set budget of \$20,000 USD, we evaluated three competing sampling schedules, with the goals of estimating both mean home range area and movement speed:

- **Option 1:** Sampling duration of **1 year**,
with a new location collected **every 15 minutes** (≈ 96 locations/day);
Model A, high cost ($\approx \$2500$ USD), tracking a maximum of 8 individuals.
- **Option 2:** Sampling duration of **6 years**,
with a new location collected **every two hours** (≈ 12 locations/day);
Model A, high cost ($\approx \$2500$ USD), tracking a maximum of 8 individuals.
- **Option 3:** Sampling duration of **4 years**,
with a new location collected **every day**;
Model B, low cost ($\leq \$1000$ USD), tracking a maximum of 20 individuals.

The relative error in population-level mean home range area and mean speed & distance estimates varied across the three competing sampling schedules (**Figure S1**). For home range estimation, **options 2 and 3** achieved a mean error within the $\pm 5\%$ threshold, with underestimations of 3.3% and 3.1%, respectively. For speed and distance estimation, **options 1 and 2** met this criterion, with their 95% confidence intervals also falling within the set threshold.

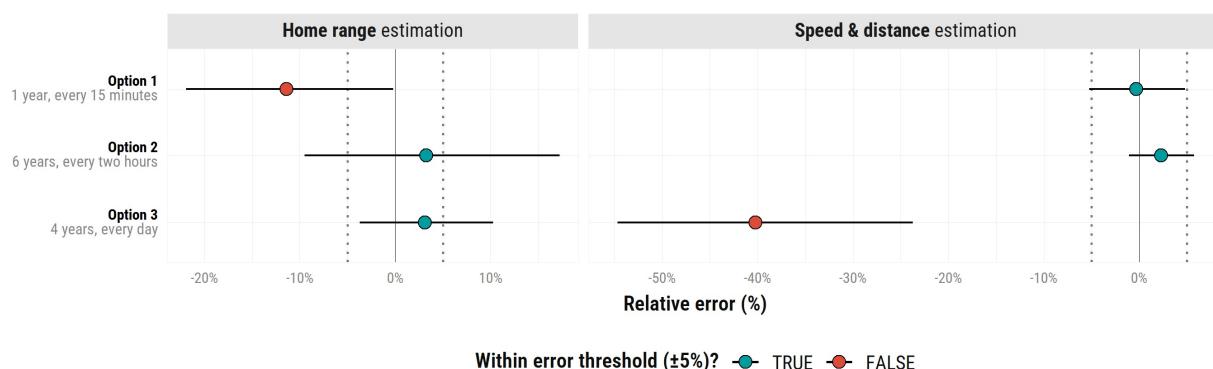


Figure S1.1. Study design comparisons for a simulated population of African buffalos (*Syncerus caffer*). Relative error (%) in home range estimation (left) and speed & distance estimation (right) for three sampling schedules. Colored points and horizontal bars represent mean error $\pm 95\%$ confidence intervals, with values falling within the error threshold in blue, and those outside of it in red. Vertical dotted lines mark the $\pm 5\%$ error threshold.

Finally, we conducted sensitivity analyses for each of our sampling schedule options. With the resampling approach for home range estimation (Figure S2, top panel), we can see that reducing the number of deployed tags further is ill-advised, as high individual variation means that lower population sample sizes have high uncertainty. For example, options 2 and 3 resulted in relative errors within the $\pm 5\%$ threshold, with underestimations of 3.3% and 3.1%, respectively. For speed and distance estimation (Figure S2, bottom panel), option 3 failed to produce reliable results, underestimating speed by over 43%. It should be noted that estimation was not possible for 17 out of 20 individuals as the data was too coarse to support a model with correlated velocity (Noonan et al., 2019). Both options 1 and 2 remained within the $\pm 5\%$ error threshold, with only a marginal difference between them, with option 2 achieving the lowest relative error (0.1%) at $m = 8$.

With our leave-one-out approach (Figure S3), and for home range estimation, option 2 exhibited greater variability in the estimate error (mean = 3.8%, SD = 2.9%) than option 3 (mean = 3.1%, SD = 0.8%). Furthermore, option 2 had a lower correctness rate (62.5%) compared to option 3. These findings highlighted that option 2 was more sensitive to each new individual included in the meta-analysis.

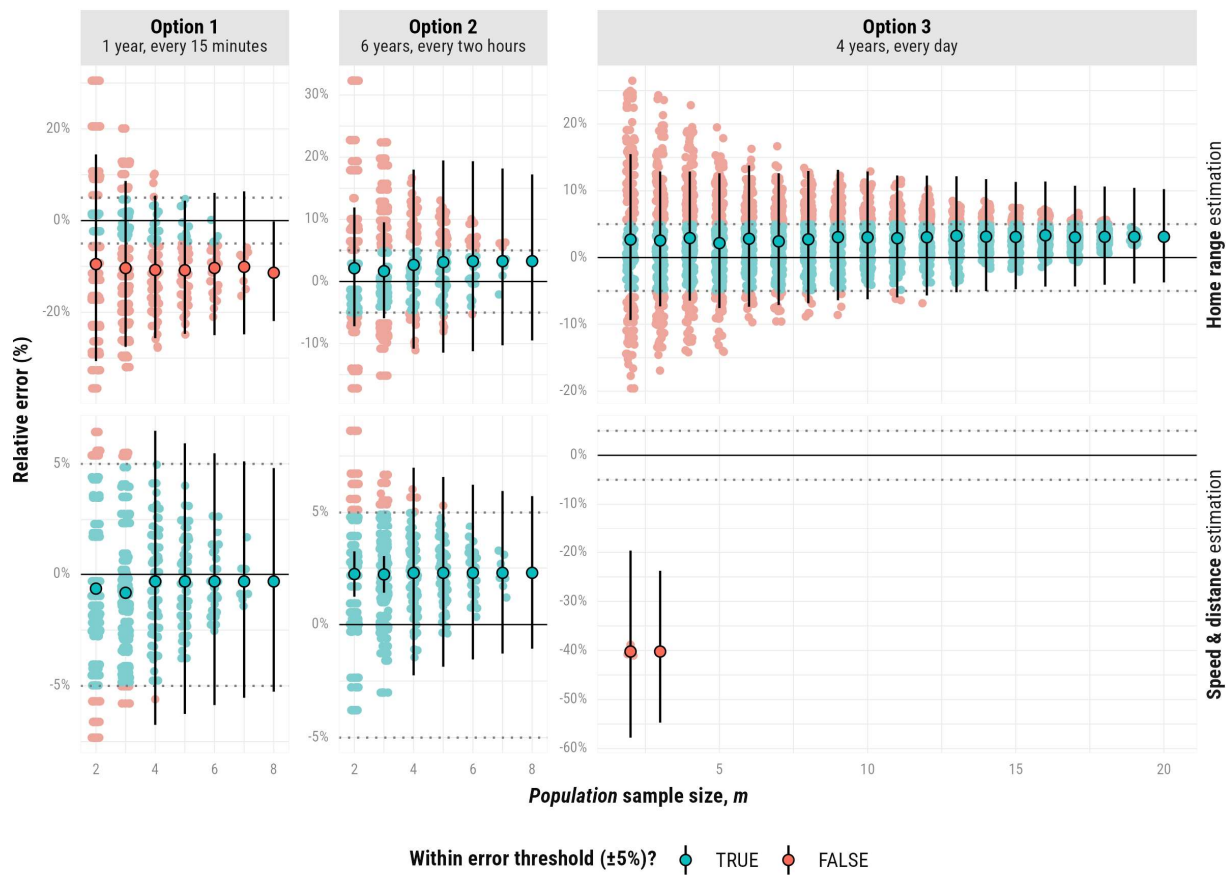


Figure S1.2. Resampling approach to assess the minimum number of tagged African buffalos (*Syncerus caffer*) needed for reliable estimates. Panels show relative error (%) in home range (AKDE, top) and speed & distance estimates (CTSD, bottom) across three sampling strategies as a function of *population* sample size (m). Horizontal dotted lines mark the $\pm 5\%$ error threshold. We generated 250 random combinations of m individuals, each represented as a colored point: in blue for estimates within the error threshold and red for estimates outside of it. Points and vertical lines in black denote the mean and confidence intervals across combinations.

Only one of our sampling schedules (**option 2**: 6-year duration, 2-hour interval, and 8 individuals) satisfied the specified error threshold for both research targets. Nevertheless, wide confidence intervals and low correctness rate suggested that a higher *population* sample size would yield more reliable estimates. These finding suggested that it may be necessary to prioritize one research target over the other, or consider alternative sampling schedules.

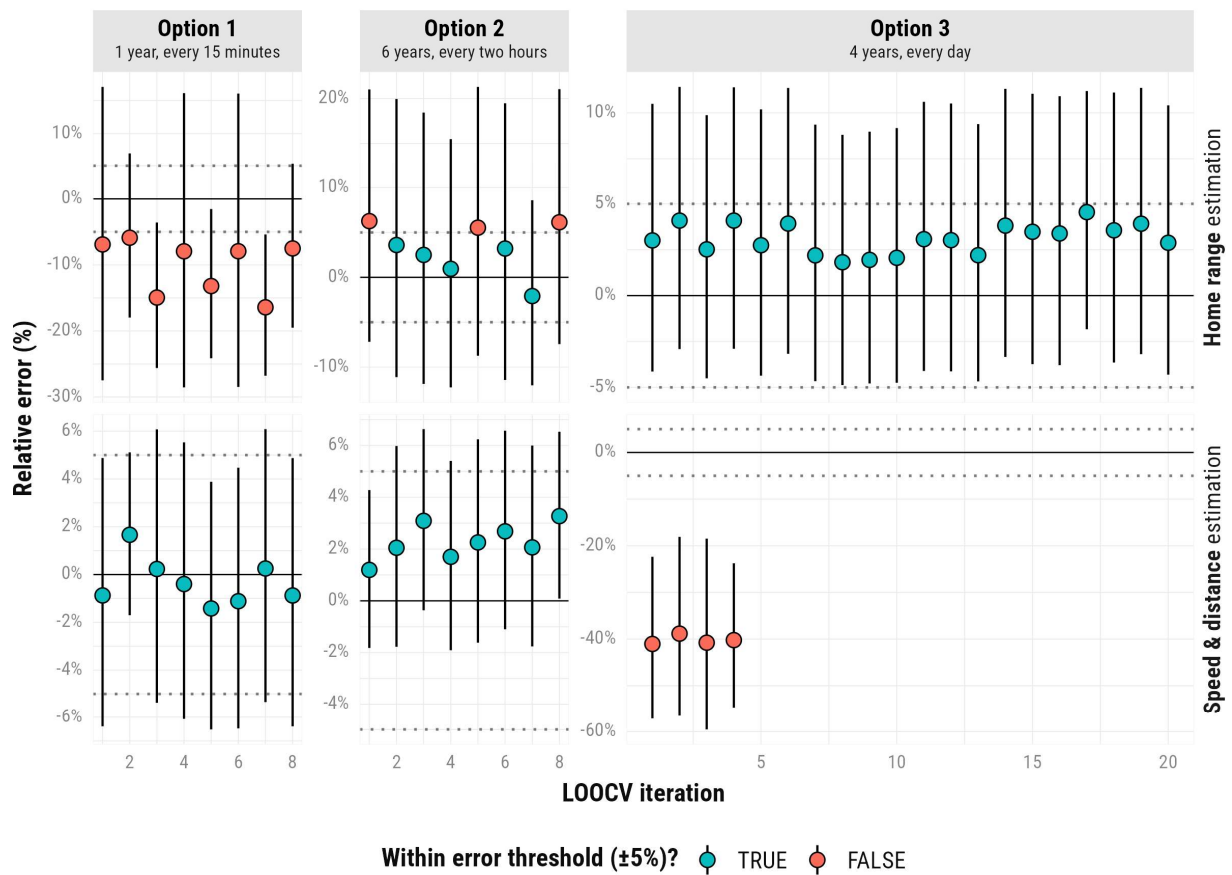


Figure S1.3. Leave-one-out approach for three competing sampling schedules. Mean relative errors (colored points) and 95% confidence intervals (bars) are shown, with values falling within the $\pm 5\%$ error threshold highlighted in blue, and those outside of it in red.

References

- Calabrese, J. M., Fleming, C. H., & Gurarie, E. (2016). ctmm: An r package for analyzing animal relocation data as a continuous-time stochastic process. *Methods in Ecology and Evolution*, 7(9), 1124–1132.
- Cross, P. C., Bowers, J. A., Hay, C. T., Wolhuter, J., Buss, P., Hofmeyr, M., Toit, J. T., & Getz, W. M. (2016). Data from: Nonparametric kernel methods for constructing home ranges and utilization distributions. Movebank data repository. <https://doi.org/10.5441/001/1.j900f88t/1>
- Fleming, C. H., Calabrese, J. M., Mueller, T., Olson, K. A., Leimgruber, P., & Fagan, W. F. (2014). From fine-scale foraging to home ranges: A semivariance approach to identifying movement modes across spatiotemporal scales. *The American Naturalist*, 183(5), E154–E167.
- Noonan, M. J., Fleming, C. H., Akre, T. S., Drescher-Lehman, J., Gurarie, E., Harrison, A.-L., Kays, R., & Calabrese, J. M. (2019). Scale-insensitive estimation of speed and distance traveled from animal tracking data. *Movement Ecology*, 7(1), 1–15.
- Silva, I., Fleming, C. H., Noonan, M. J., Alston, J., Folta, C., Fagan, W. F., & Calabrese, J. M. (2022). Autocorrelation-informed home range estimation: A review and practical guide. *Methods in Ecology and Evolution*, 13(3), 534–544.